paths.py — technical

`src/moe_pruning/utils/paths.py · 134 lines · moe_pruning.utils.paths`

Purpose

Artifact layout and JSON/hash helpers.

Data and structure

- `<EXPERIMENT_DIR>/<EXPERIMENT_NAME>/`: architecture_report.json, expert_scores.*, expert_selection.json, bias_initialization.json, replaced-model/, trained-model/, training_report.json, evaluation/, config/, experiment.log.
- `<STATS_DIR>/<EXPERIMENT_NAME>/`: expert_stats.pt, expert_stats_manifest.json.
  `<CACHE_DIR>/<EXPERIMENT_NAME>/`: eval_examples.pt, layer_reference/.

Step by step

Classes and functions in source order. Signatures, line ranges, raised exceptions and settings are extracted from the code; numbered steps describe what the code does.

`class MissingArtifactError`

lines 17-18 · bases FileNotFoundError

Raised when a stage depends on an artifact that has not been produced yet.

`class ArtifactPaths`

lines 21-108

Locations of all artifacts of the configured experiment.

`ArtifactPaths.architecture_report`

```
def architecture_report(self) -> Path
```

lines 27-28 · settings EXPERIMENT_DIR/EXPERIMENT_NAME

Location of the architecture report artifact.

`ArtifactPaths.expert_stats`

```
def expert_stats(self) -> Path
```

lines 32-33 · settings STATS_DIR/EXPERIMENT_NAME

Location of the expert stats artifact.

`ArtifactPaths.expert_stats_manifest`

```
def expert_stats_manifest(self) -> Path
```

lines 36-37 · settings STATS_DIR/EXPERIMENT_NAME

Location of the expert stats manifest artifact.

`ArtifactPaths.expert_scores_json`

```
def expert_scores_json(self) -> Path
```

lines 40-41 · settings EXPERIMENT_DIR/EXPERIMENT_NAME

Location of the expert scores json artifact.

`ArtifactPaths.expert_scores_csv`

```
def expert_scores_csv(self) -> Path
```

lines 44-45 · settings EXPERIMENT_DIR/EXPERIMENT_NAME

Location of the expert scores csv artifact.

ArtifactPaths.expert_selection

```
def expert_selection(self) -> Path
```

lines 48-49 · settings EXPERIMENT_DIR/EXPERIMENT_NAME

Location of the expert selection artifact.

ArtifactPaths.replaced_model_dir

```
def replaced_model_dir(self) -> Path
```

lines 53-54 · settings EXPERIMENT_DIR/EXPERIMENT_NAME

Location of the replaced model dir artifact.

ArtifactPaths.trained_model_dir

```
def trained_model_dir(self) -> Path
```

lines 57-58 · settings EXPERIMENT_DIR/EXPERIMENT_NAME

Location of the trained model dir artifact.

ArtifactPaths.bias_initialization_report

```
def bias_initialization_report(self) -> Path
```

lines 61-62 · settings EXPERIMENT_DIR/EXPERIMENT_NAME

Location of the bias initialization report artifact.

ArtifactPaths.layer_cache_dir

```
def layer_cache_dir(self) -> Path
```

lines 66-67 · settings CACHE_DIR/EXPERIMENT_NAME

Location of the layer cache dir artifact.

ArtifactPaths.training_report

```
def training_report(self) -> Path
```

lines 70-71 · settings EXPERIMENT_DIR/EXPERIMENT_NAME

Location of the training report artifact.

ArtifactPaths.eval_dir

```
def eval_dir(self) -> Path
```

lines 75-76 · settings EXPERIMENT_DIR/EXPERIMENT_NAME

Location of the eval dir artifact.

ArtifactPaths.eval_metrics

```
def eval_metrics(self, stage: str) -> Path
```

lines 78-79

Metrics file of an evaluation stage.

ArtifactPaths.eval_per_example

```
def eval_per_example(self, stage: str) -> Path
```

lines 81-82

Per-example results of an evaluation stage.

ArtifactPaths.eval_comparison_json

```
def eval_comparison_json(self) -> Path
```

lines 85-86

Location of the eval comparison json artifact.

`ArtifactPaths.eval_comparison_md`

```
def eval_comparison_md(self) -> Path
```

lines 89-90

Location of the eval comparison md artifact.

`ArtifactPaths.eval_examples_cache`

```
def eval_examples_cache(self) -> Path
```

lines 93-94 · settings CACHE_DIR/EXPERIMENT_NAME

Location of the eval examples cache artifact.

`ArtifactPaths.log_file`

```
def log_file(self) -> Path
```

lines 98-99 · settings EXPERIMENT_DIR/EXPERIMENT_NAME

Location of the log file artifact.

`ArtifactPaths.config_snapshot_dir`

```
def config_snapshot_dir(self) -> Path
```

lines 102-103 · settings EXPERIMENT_DIR/EXPERIMENT_NAME

Location of the config snapshot dir artifact.

`ArtifactPaths.require`

```
def require(self, path: Path, produced_by: str) -> Path
```

lines 105-108 · raises MissingArtifactError

Return the path, or stop with the command that produces it.

`write_json`

```
def write_json(path: Path, data: Any) -> Path
```

lines 111-116

Write JSON atomically (temporary file, then rename).

1. Create parent dirs; write to `<name>.tmp`; atomic `replace` - readers never see half-written files.

`read_json`

```
def read_json(path: Path) -> Any
```

lines 119-120

Read a JSON file.

`file_sha256`

```
def file_sha256(path: Path, chunk_size: int=1 << 20) -> str
```

lines 123-128

SHA-256 of a file (identifies datasets in fingerprints).

`fingerprint`

```
def fingerprint(data: Any) -> str
```

lines 131-134

Stable short hash of a JSON-serialisable object.

1. `json.dumps(sort_keys=True)` then SHA-256, first 16 hex characters.

Errors raised

Function	Exception	Message (f-string fields in braces)
`require`	MissingArtifactError	Required artifact {path} not found. Run `{produced_by}` first.